MILITARY SPECIFICATION SHEET

SCANNER, MECHANICAL, MX-10717(V)5/U

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1. SCOPE

1.1 Scope. This specification sheet covers the Mechanical Scanner, MX-10717(V)5/U and augments MIL-S-49301. The MX-10717(V)5/U is used on the M1 Tank Thermal Imaging System.

1.2 Purpose. The purpose of this specification sheet is to modify or expand the contents of MIL-S-49301 to describe the MX-10717(V)5/U Mechanical Scanner. All the MIL-S-49301 contents apply except as specifically modified herein. Paragraph numbers directly correlate between MIL-S-49301 and this specification sheet.

2. APPLICABLE DOCUMENTS

2.1 Order of precedence. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.

SPECIFICATIONS

MILITARY

MIL-S-49301	- Scanner, Mechanical, MX-10717(V)/U, General Specification for

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

80063-SM-D-773946	- Scan-Interlace, PL-1408/UA
54490-5006243-5	- Scanner, Mechanical, MX-10717(V)5/U

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 Description. The MX-10717(V)5/U Scanner, herein known as the MX-10717(V)5/U, shall consist of the MIL-S-49301, MX-10717(V)/U Scanner adjusted for a nominal 40.3 ±1.0 Hz operation and 12.6 ± 0.2 degrees total scan angle.

3.2 Construction. The MX-10717(V)5/U shall be constructed in accordance with drawing 5006243-5.

3.2.1 Weight. Weight of the MX-10717(V)5/U shall be 1.90 pounds maximum.

3.6 Performance characteristics. The characteristics specified herein shall apply as constructed (3.2). Operation shall be at a nominal 40.3 ±1.0 Hz, 2:1 side interlace mode and a spring limit pulse of 2.7 ± 0.1 ms while using the scan-interlace, PL-1408/UA (SM-D-773946).

3.6.1 Scan angle and frequency.

3.6.1.1 Variable and scan frequency. Applicable for first article test only. See (MIL-S-49301.)

3.6.1.2 Specific scan frequency.(40.3 ± 1.0 Hz) Scan frequency of the MX-10717(V)5/U shall be adjusted 40.3 ± 1.0 Hz. Scan frequency drift shall not be greater than ± 2.0 Hz from the frequency at ambient setting (see MIL-S-49301, 4.4) when exposed to high and low temperature environmental conditions (MIL-S-49301).

3.6.1.3 Synchronous scan frequency. Not applicable.

3.6.1.4 Scan angle. The scan angle shall be 6.3 ± 0.1 degrees in each direction (12.6 ± 0.2 degrees total) from a true 45-degree position with respect to the mounting surface (see figure 1).

3.6.1.5 Scan transducer D.C. offset. The scan position transducer shall have a D.C. offset of 0.0 ± 0.2 Vdc.

3.6.2 Interlace. The MX-10717(V)5/U shall operate in a 2:1 side interlace mode. The switching time shown in figure 1 of MIL-S-49301 shall not be greater than 2.7 milliseconds (ms) and the average amplitude shall be 1.0 ± 0.2 Vp-p.

3.6.2.1 Interlace angle. The interlace solenoid stops and scan-interlace, PL-1408/UA (SM-D-773946) potentiometers shall be used to determine a dynamic object space interlace angle of 155 ± 15 arc seconds.

3.6.2.2 Interlace solenoid position. The distance of the interlace solenoid assembly from the solenoid plungers, herein defined as the air gap, shall be equivalent to a 1.0 ± 0.2 Vdc swing from a mechanical gimbal zero position, when measured at J-6 of the scan-interlace, PL-1408/UA (SM-D-773946).

3.6.2.3 Elevation boresight. The scan mirror shall not deviate by more than ± 16 arc minutes from the scanner Datum-A- (Datum -A- is the optical axis and is parallel to the scanner mounting surface, see Fig. 2).

3.6.2.4 Interlace transducer D.C. offset. The interlace position transducer shall have a D.C. offset of 0.0 ± 0.2 Vdc.

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3.6.3 Scan direction. The MX-10717(V)5/U shall show the waveform shapes and phase relationships in figure 3 while operating in the 2:1 side interlace mode.

3.6.4 Power consumption. Power required to operate the MX-10717(V)5/U connected to the Scan-interlace, PL-1408/UA, shall not exceed 6.3 watts total. Distribution of power shall be as follows:

a.	+ 5.0 ± 0.3 Vdc	≤	0.65 watts
b.	- 5.0 + 0.3 Vdc	≤	0.25 watts
c.	+15.0 ± 0.5 Vdc	≤	2.70 watts*
d.	-15.0 ± 0.5 Vdc	≤	2.70 watts*

*Note: The +15 and -15 Vdc power shall have a difference of ≤ 0.6 watts.

3.6.5 Scan efficiency. The MX-10717(V)5/U scan efficiency shall be a minimum of 65 percent.

3.6.6 Scan jitter in azimuth. Jitter time shall be 18.0 microseconds maximum.

3.6.7 Scan Position. The scan position waveform shown in figure 3 shall have an amplitude of 1.0 ± 0.1 Vp-p when at J3 of the Scan Interlace, PL-1408/UA (SM-D-773946).

3.7 Environmental conditions. See MIL-S-49301 except as follows:

3.7.5 Vibration. The MX-10717(V)5/U shall not be damaged by vibration over a frequency spectrum at the specified g levels and amplitudes shown in figure 4 for the X axis and as shown in figure 4 of MIL-S-49301 for the Y and Z axes. During specified vibration, the average interlace waveform amplitude shown in MIL-S-49301, figure 1, shall not deviate from the average amplitude without vibration by more than ± 50 percent.

4. QUALITY ASSURANCE PROVISIONS

4.3 First article inspection. See MIL-S-49301.

4.3.2 Test. In addition to the requirements of MIL-S-49301, table II, the requirements of 3.6.1.5, 3.6.2.1, 3.6.2.2, 3.6.2.3 and 3.6.2.4 shall also be met. Failure of any test shall constitute failure of first article inspection.

4.5 Quality conformance inspection.

4.5.2 Tests.

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4.5.2.1 Group A inspection. In addition to the requirements of MIL-S-49301 table III, the requirements of paragraphs 3.6.1.5, 3.6.2.1, 3.6.2.2, 3.6.2.3, 3.6.2.4 and test paragraphs 4.6.1.5, 4.6.2.1, 4.6.2.2, 4.6.2.3, and 4.6.2.4 shall also be met. Failure of any test shall be cause for rejection of the unit.

4.6 Test methods. Tests shall be performed utilizing the SM-D-773946 Scan-Interlace with the MX-10717(V)5/U.

4.6.1 Scan frequency.

4.6.1.1 Variable scan frequency. See MIL-S-49301.

4.6.1.2 Specific scan frequency (40.3 Hz). The MX-10717(V)5/U shall be connected to the Scan-Interlace SM-D-773946, and mounted to a suitable fixture. Power shall be applied and the output of the azimuth (scan) transducer (Scan-Interlace SM-D-773946) shall be monitored with a calibrated frequency counter. The output of the Scan-Interlace (SM-D-773946) shall be adjusted to obtain the specified frequency. Failure to meet the requirements of 3.6.1.2 shall constitute failure of this test.

4.6.1.3 Synchronous scan frequency. Not applicable.

4.6.1.4 Scan angle. Scan angle measurement shall be performed in accordance with MIL-S-49301. Scan angle shall be accordance with 3.6.1.4.

4.6.1.5 Scan transducer D.C. offset. The D.C. offset of the scan position transducer shall be measured at J3 of the scan interlace, PL-1408/UA (SM-D-773946) with the scan mirror in the mechanical zero position. The D.C. offset shall meet the requirements of 3.6.1.5.

4.6.2 Interlace. With the scanner mounted in a suitable fixture, the output of the interlace position transducer shall be monitored with an oscilloscope. The waveform switching times (figure 1 of MIL-S-49301) and amplitude shall be measured to verify that the requirements of 3.6.2 are met. Failure to meet the requirements of 3.6.2 shall constitute failure of this test.

4.6.2.1 Interlace angle. Using an autocollimator and an auxiliary mirror to measure along DATUM -A- (DATUM -A- is the optical axis), the interlace solenoid stops and Scan-Interlace, PL-1408/UA, shall be adjusted to provide a dynamic object space interlace angle of 155 $\pm$ 15 arc seconds. Failure to meet the requirements of 3.6.2.1 shall constitute a failure of this test.

4.6.2.2 Interlace solenoid position. With the scanner mounted to a suitable fixture and the scanner disabled, the offset from the interlace solenoid assembly to the plunger shall be measured at J6 of the scan-interlace, PL-1408/UA. The offset shall be equivalent to + 1.0 $\pm$ 0.2 Vdc from a mechanical gimbal zero position. The interlace position transducer (MT2) is to have a target to transducer distance of 0.025 $\pm$ 0.005 inches, and a transducer bridge voltage of 1.90 $\pm$ 0.1 Vdc. The interlace solenoid stops are to be retracted so the interlace plungers make contact with the interlace solenoid assembly. Failure to meet the requirements of 3.6.2.2 shall constitute a failure of this test.

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4.6.2.3 Elevation boresight. The scan mirror shall not deviate by more than ± 16 arc minutes in elevation from the scanner DATUM -A- (DATUM -A- is the optical axis) (see figure 2). Elevation boresight shall be measured at the approximate center of the azimuth scan travel with interlace not operating. Failure to meet the requirements of 3.6.2.3 shall constitute a failure of this test.

4.6.2.4 Interlace tranducer D.C. offset. The D.C. offset of the interlace position transducer shall be measured at J6 of the scan interlace, PL-1408/UA (SM-D-773946), with the interlace gimbal in the mechanical zero position. The D.C. offset shall meet the requirements of 3.6.2.4. Failure to meet the requirements of 3.6.2.4 shall constitute a failure of this test.

4.6.3 Scan direction. With the MX-10717(V)5/U operating in the 2:1 side interlace mode, the outputs of the scan (J3) and interlace (J6) position transducers (Scan-Interlace, PL-1408/UA, SM-D-773946) shall be observed with an oscilloscope. The output waveforms shall show the shape and phase relationships shown in figure 3. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Power consumption. See MIL-S-49301.

4.6.5 Scan efficiency. MIL-S-49301.

4.6.6 Scan jitter in azimuth. See MIL-S-49301.

4.6.7 Scan Position. With the scanner mounted in a suitable fixture, the output at the scan position transducer (MT1) shall be monitored with an oscilloscope. The amplitude shall be measured to verify that the requirements of 3.6.7 are met. Failure to meet the requirements of 3.6.7 shall constitute failure of this test.

4.7 Environmental tests. See MIL-S-49301 and figure 4.

6. NOTES

6.1 Intended use. The MX-10717(V)5/U is intended for use with the Thermal Imaging System of the M1 Tank.

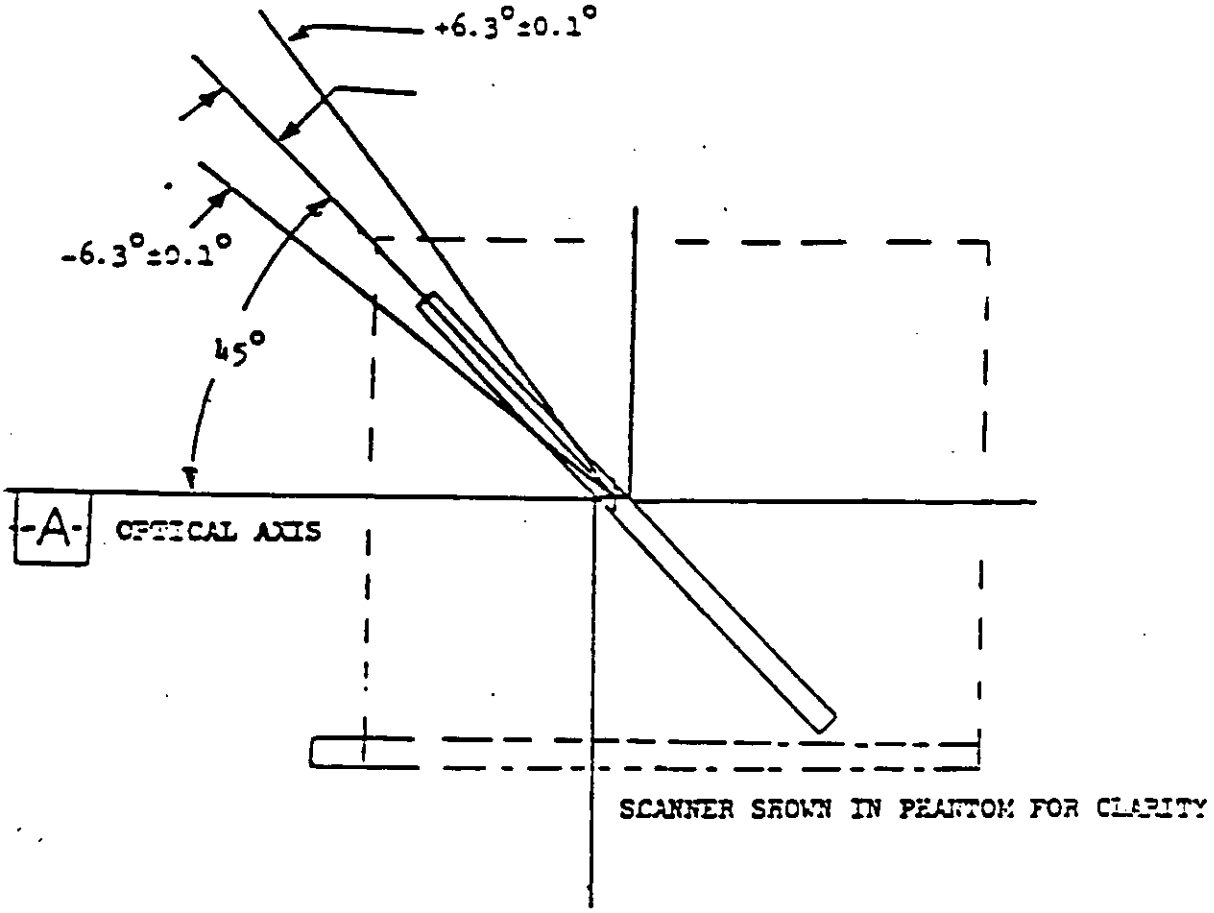
6.2 Interchangeability. The MX-10717(V)5/U scanner built and tested in accordance with this specification and applicable drawings may be used in place of the MX-9872(V)5/UA scanner. The MX-9872(V)5/UA may not be used in place of the MX-10717(V)5/U scanner.

6.3 Subject Term (Keyword) Listing.
FLIR
Sensor
Scanner

CUSTODIAN:
Army - CR

PREPARING ACTIVITY
Army - CR
Project 5855-A290-5

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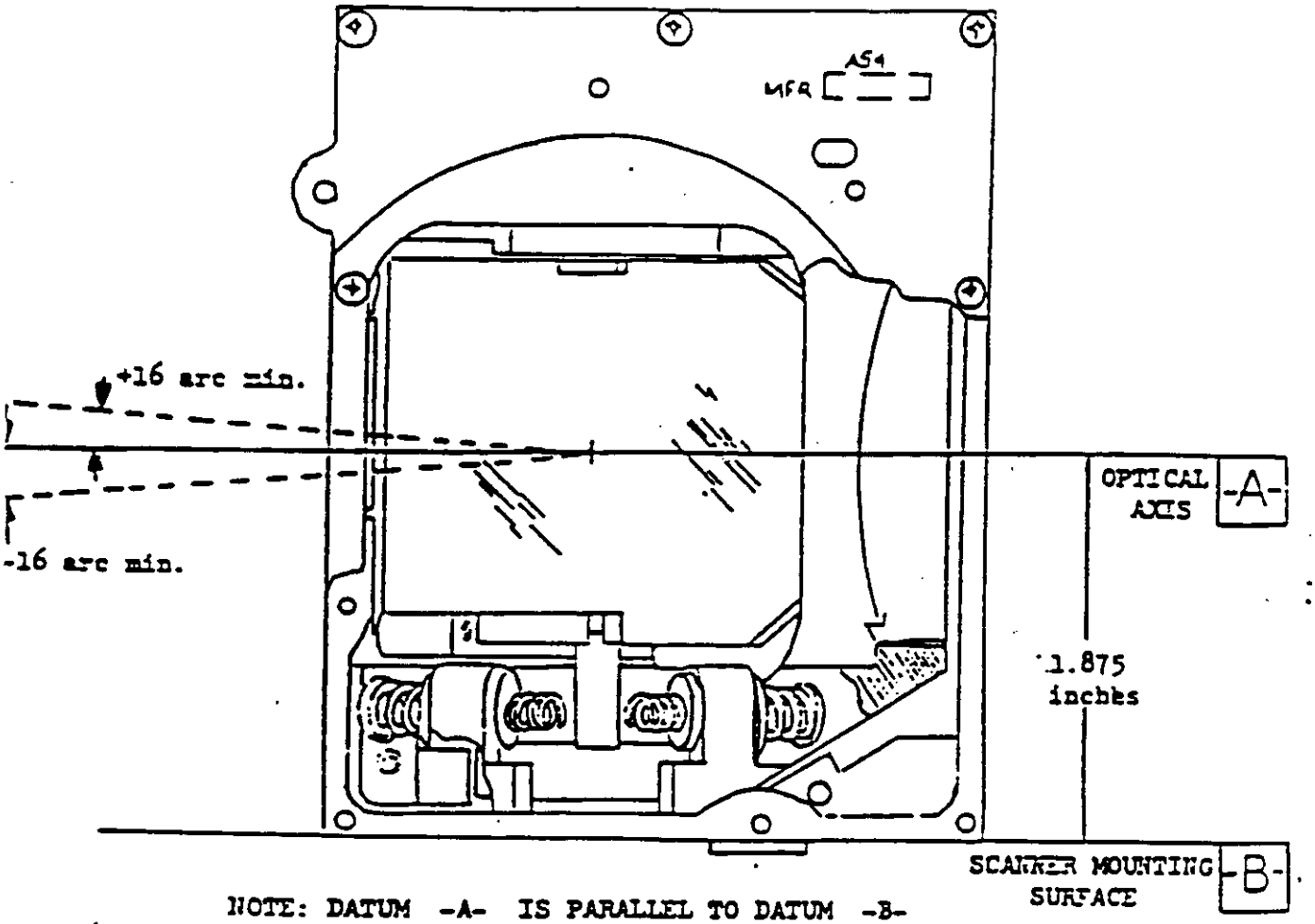


THE AZIMUTH SCAN ANGLE IS DYNAMICALLY SET FOR THE TOTAL SCAN ANGLE OF 12.6°±0.2°. SCAN ANGLE IS TO BE +6.3°±0.1° AND -6.3°±0.1° FROM THE MIRROR POSITION ORIENTED AT 45° FROM THE SCANNER DATUM -A-.

DATUM -A- THE OPTICAL AXIS IS DEFINED TO BE PARALLEL TO THE MOUNTING SURFACE OF THE SCANNER.

FIGURE. 1 SCAN ANGLE.

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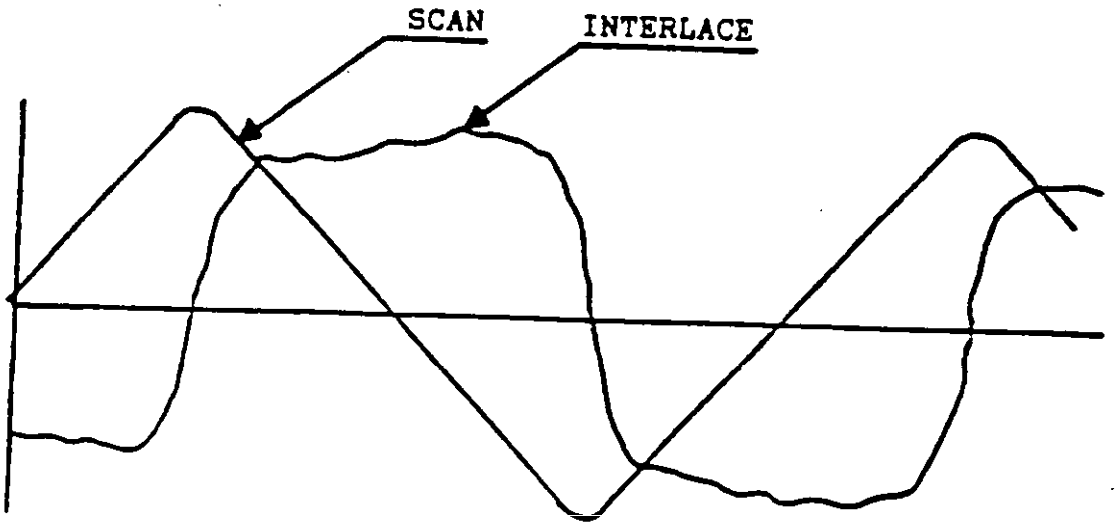
NOTE: DATUM -A- IS PARALLEL TO DATUM -B-

FIGURE 2. ELEVATION BORESIGHT.

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2:1 SIDE POSITION

INTERLACE FOLLOWS SCAN BY APPROX. 90°



NOTES:

1. PL-1408(V)/UA: CONNECT P1-31 TO P1-33 AND P1-4 TO P1-17.
2. OSCILLOSCOPE: CHANNEL 1 - HIGH INPUT TO J3 (OR J2-7), LOW TO GND.

   CHANNEL 2 - HIGH INPUT TO J6 (OR J1-9), LOW TO GND.

FIGURE 3. INTERLACE TRANSDUCER AND AZIMUTH POSITION TRANSDUCER OUTPUTS.

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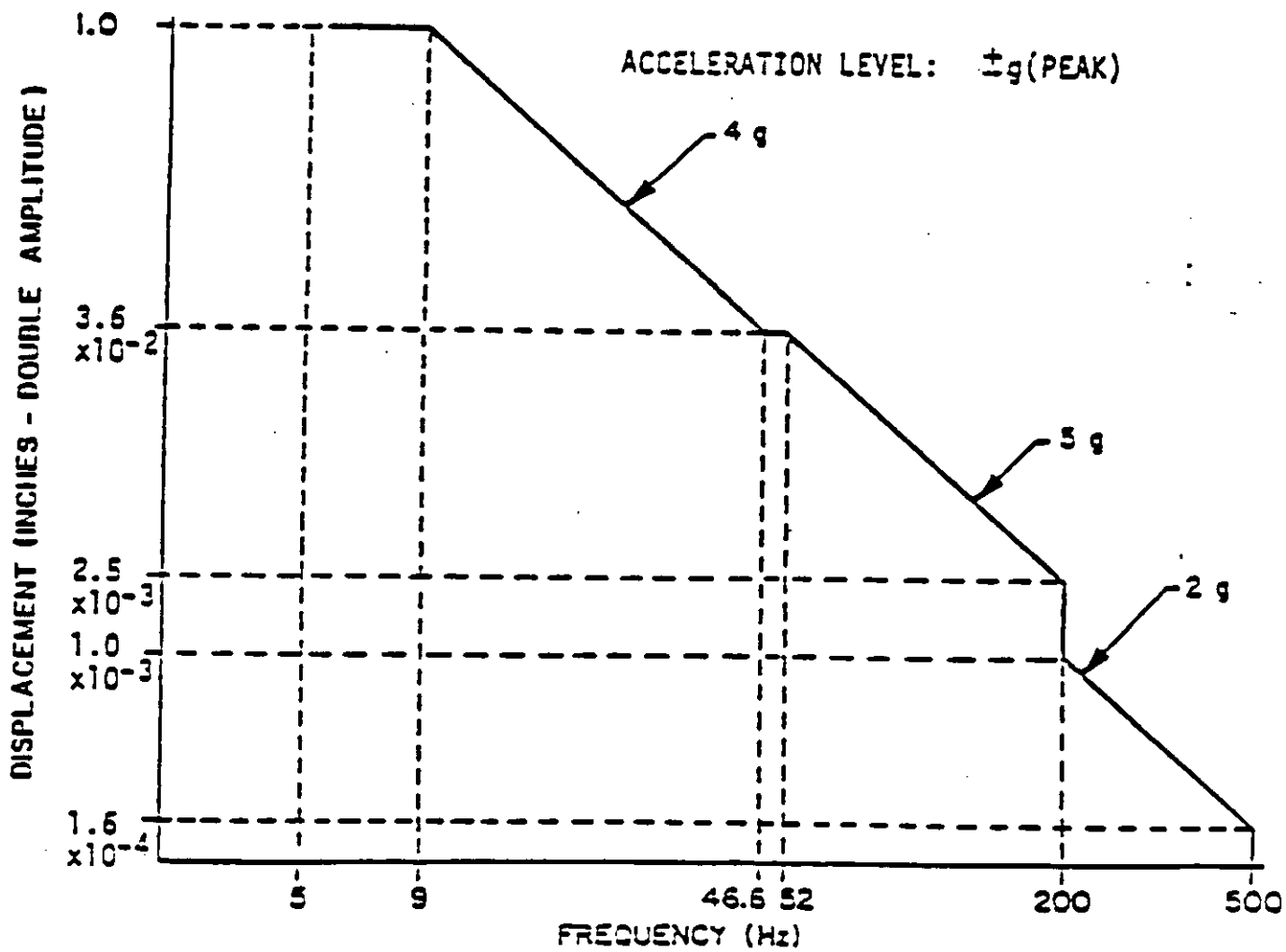


FIGURE 4. VIBRATION TEST PROFILE (X AXIS).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-49301/5(CR)	Scanner, Mechanical, MX-10717(V)5/U

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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